Occlusion Detection

August to September 2026

The goal was to place an ad inside a manually selected region of interest while keeping foreground objects in front of it. When an object crossed the region, the corresponding ad pixels had to be hidden. A green polygon was used as the test ad so that the occlusion mask could be evaluated separately from ad design.

Assumptions

Each experiment used the first video frame as a fixed background reference. Four manually selected points defined the region of interest. The camera and selected surface were assumed to remain static, so later frames were not aligned and the region was not tracked. Some test regions were selected to exercise difficult cases rather than to show ideal ad placement.

The methods progressed from direct frame difference to LAB chroma difference, optical flow, and edge-based segmentation.

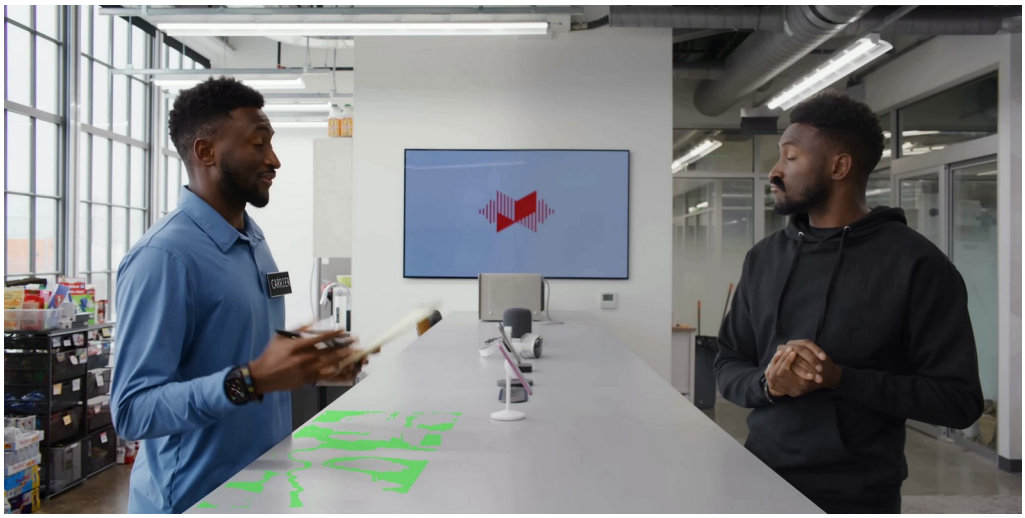
Original code

The first prototype calculated their absolute BGR difference. The mask was inverted so that 255 meant the ad was visible and 0 meant it was occluded.

The method detects a change in pixel values; it does not recognise the object or determine whether that change is actually in front of the ad. Even small changes in brightness can therefore hide the ad. Moving shadows, reflections, exposure changes, or slight camera movement can all cause this pattern.




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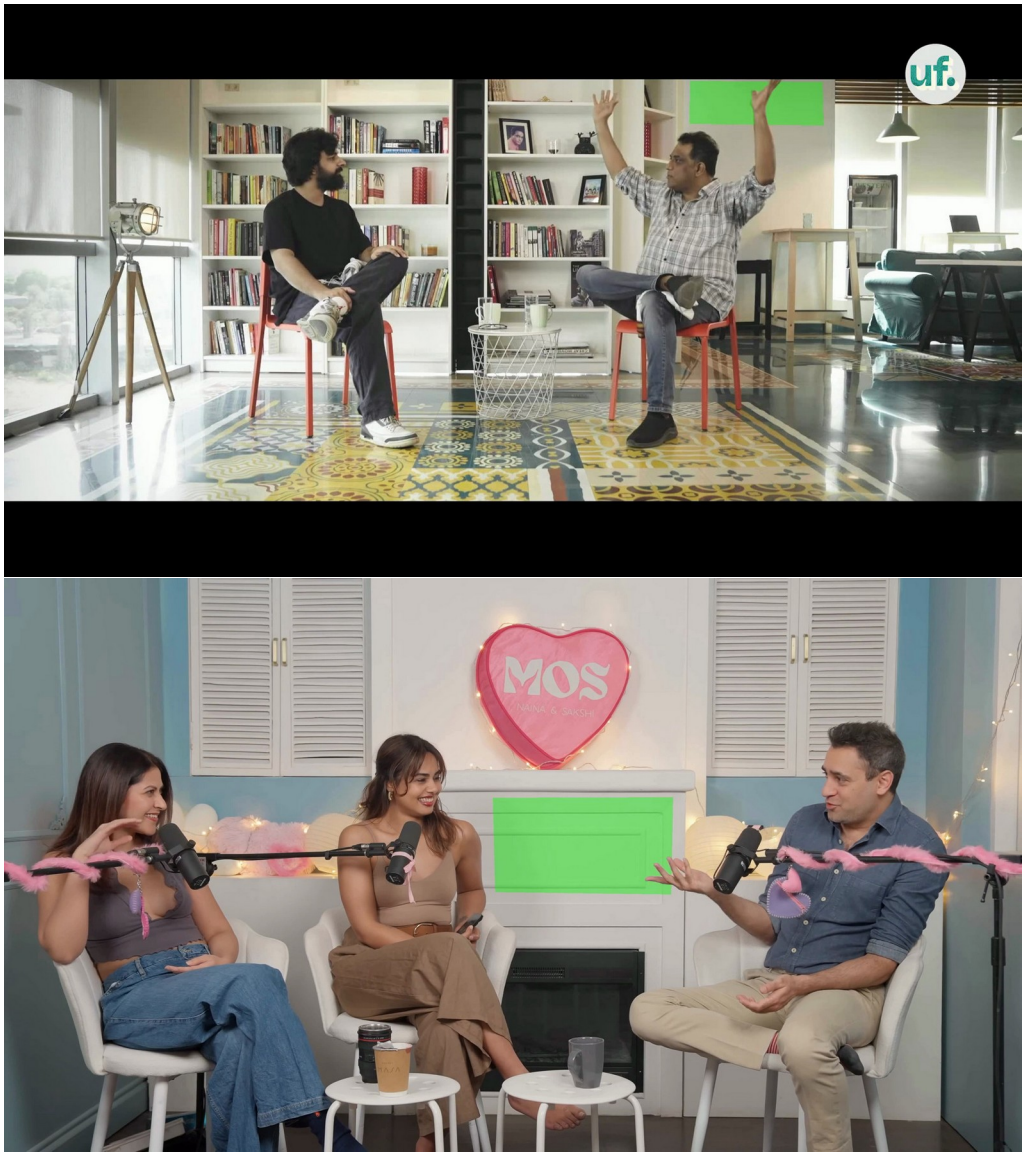
First Idea: Lighting

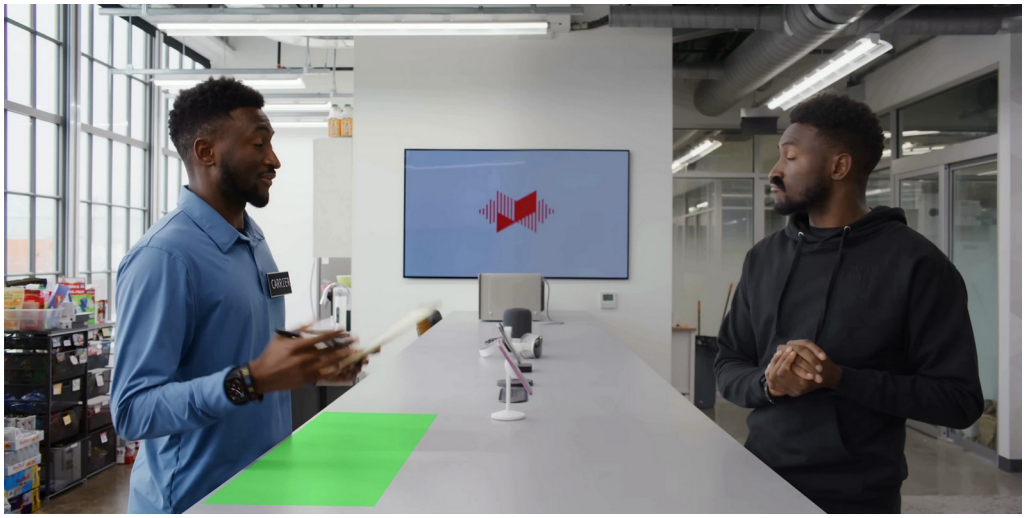
The lighting methods were intended to reduce the false occlusion caused by brightness changes in the original code. The main change is the colour information used to decide whether a pixel belongs to an occluder.

We first convert the reference and current crops to LAB, and compare the A and B colour channels. L represents lightness, while A and B describe colour along the green-red and blue-yellow axes.

Ignoring L means that a surface can become brighter or darker without automatically being classified as an occluder, provided its A and B values remain similar. This can preserve the ad under a neutral shadow more effectively than raw BGR subtraction. It is not complete shadow removal: coloured lighting, reflections, camera colour adjustments, and some shadows also change A or B and can still create holes in the green overlay.

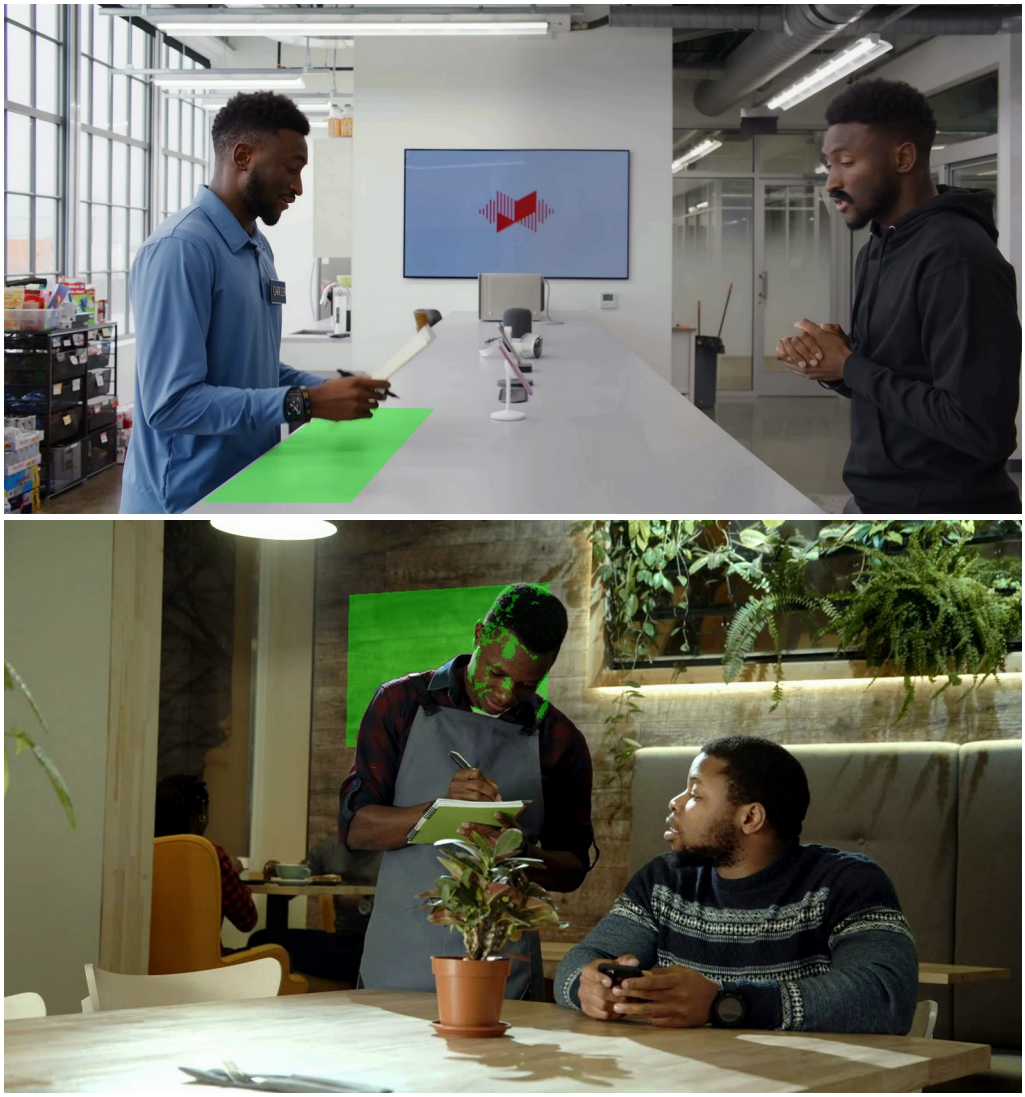
This method is consistent with suppressing brightness-only changes while retaining stronger colour differences. But these pixels have not produced enough chroma change to hide the ad. Background preservation alone is therefore insufficient; foreground completeness must also be checked.








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Second Idea: Optical Flow

The optical-flow branch used dense Farneback flow between consecutive grayscale frames. Flow did not create or move the foreground mask. LAB chroma difference first produced connected foreground candidates, and one component was selected and matched across frames using overlap and centroid distance. The median flow magnitude helped label the component as outside, entering, or inside the region. The track was kept for up to three missed frames.

This added short-term continuity and prevented every changed region from hiding the ad. It still tracked only one object, inherited gaps from the colour mask, and could be disturbed by camera motion. Because flow was used only as a state signal, it did not recover missing object pixels or propagate the mask.

Third Idea: Edge Detection

The edge idea adds boundary and region information to the colour-change detector. The five versions differ in whether edges create foreground, merely support colour seeds, or refine an existing segmentation.

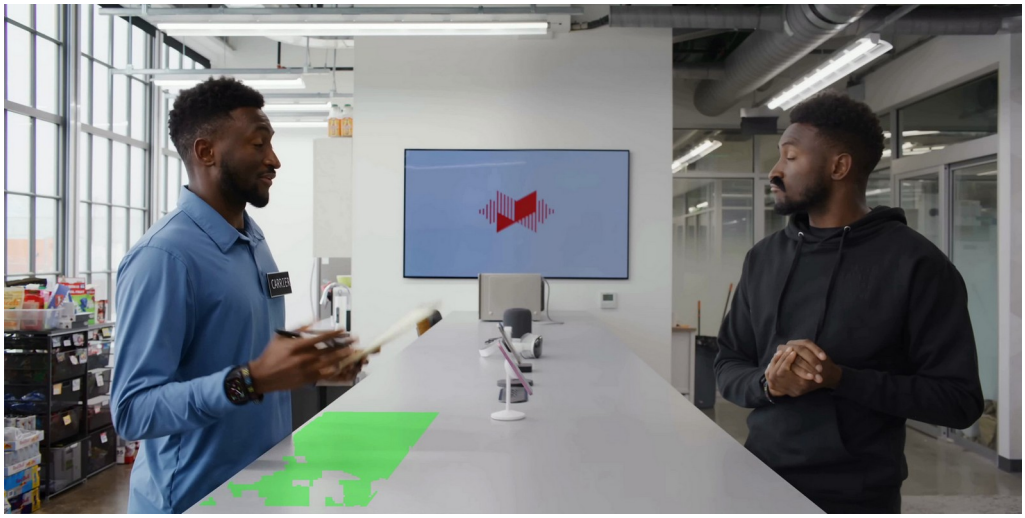
Version 1

The first version analyses a crop padded by 10 pixels. It converts both frames to LAB, applies a 3 by 3 box blur, and averages the absolute A and B differences. Unlike the lighting version, it accepts very small chroma variations rather than requiring a threshold value as in the lighting code.

Canny uses thresholds of 50 and 150 on the current blurred LAB image. This is an edge map of the current scene. An OR operation combines it with the colour-change mask. External contours larger than one per cent of the crop area are filled; smaller contours are discarded.

Filling contours can turn a fragmented outline into a solid occluder, but it also fills outlines belonging to the background. Static furniture, wall detail, shadows with sharp boundaries, and low-level colour noise can enter the combined mask. Closing may connect those detections into a large region, and the area filter then preserves that region. Conversely, a small or narrow real object may fail the area test.






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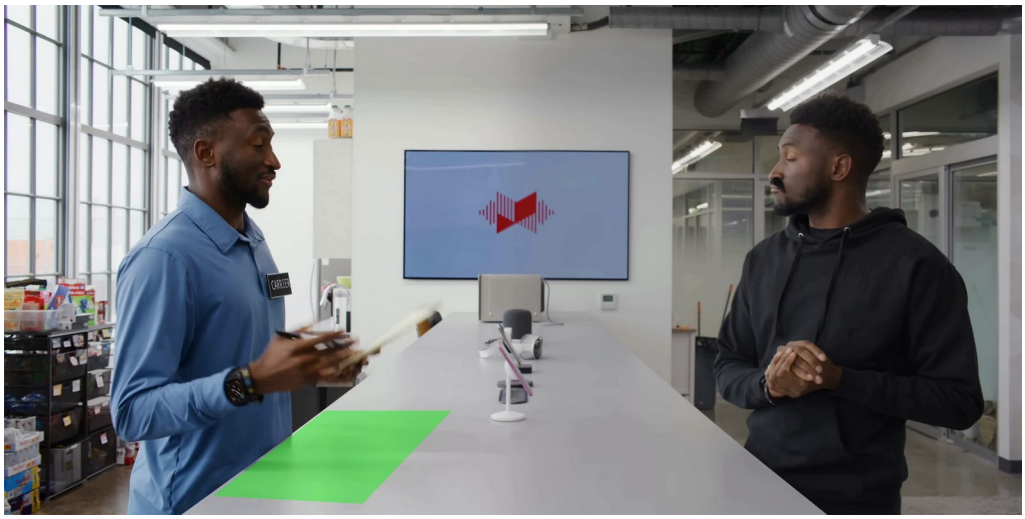
Version 2

This code does edge selection depend on colour change. Canny is applied separately to the current A and B channels and the two edge maps are combined. Unlike Version 1, this version does not fill all external contours or remove components using a crop-relative area threshold.

This restriction stops an unrelated background outline from creating foreground by itself. Excluding lightness from both the difference and the edge channels also reduces sensitivity to neutral illumination changes. However, edges can only support nearby seeds. A grey object against a grey background may produce neither a strong chroma difference nor a chroma edge, leaving green on the object.


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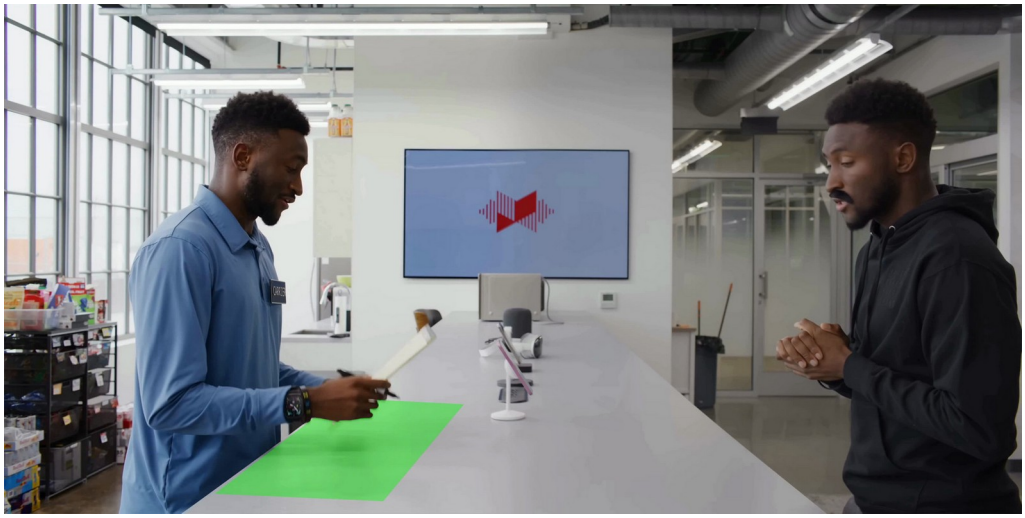

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Version 3

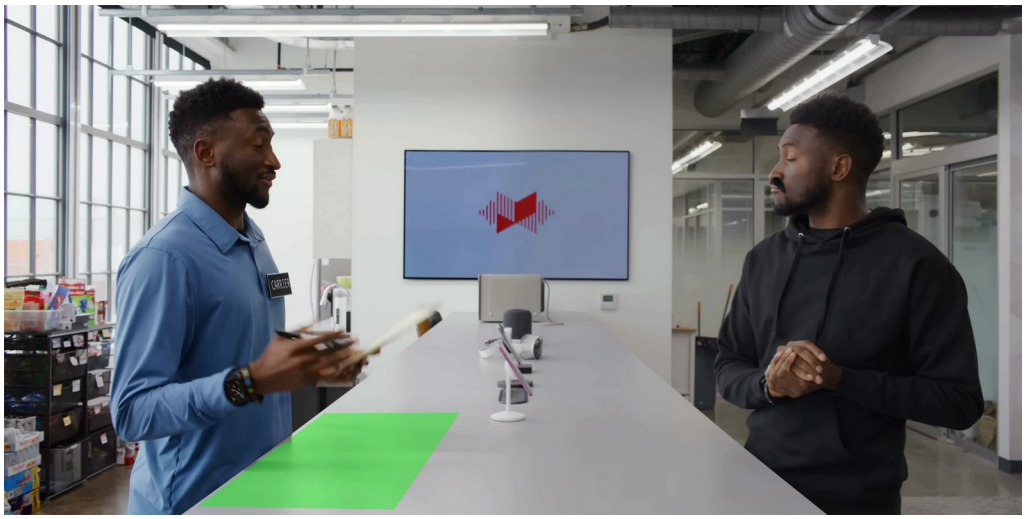
Version 3 introduces GrabCut to estimate a foreground region from initial labels. Separately, grayscale Canny edges use thresholds of 50 and 150 to produce a rough silhouette.

The edge silhouette is labelled probable foreground, while the colour-change pixels are definite foreground. All remaining pixels start as probable background, except the crop border, which is forced to definite backgroundIt can revise probable labels according to foreground and background appearance, but cannot reject definite foreground seeds. If usable foreground and background labels are not both present, the code returns no detected occlusion.

Grayscale edges can suggest a neutral object even when chroma change is weak, but they also describe static objects and illumination boundaries. The small crop provides limited background context, and forcing its border to background can truncate an object entering through that border.


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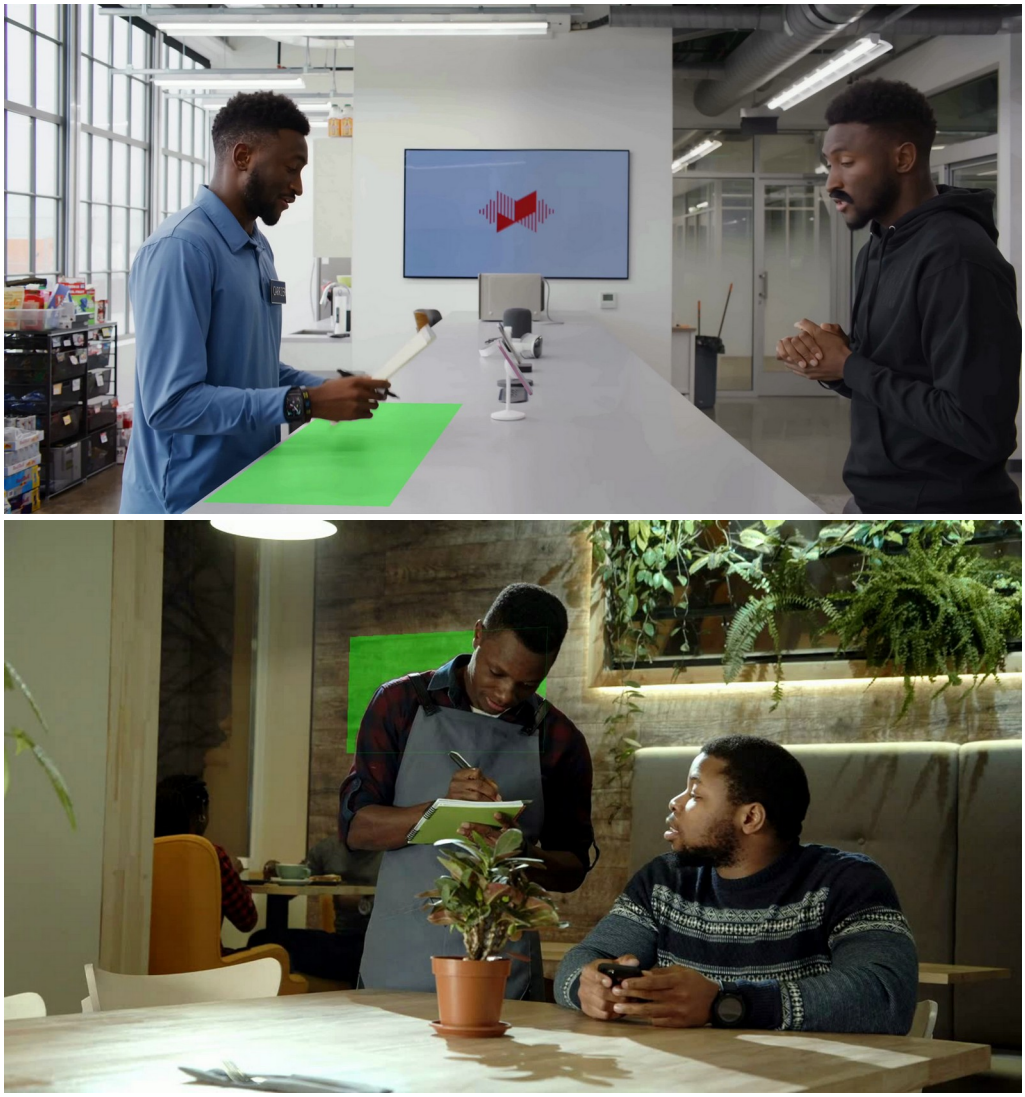

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Version 4

Version 4 uses a 20-pixel padded crop and changes the order: colour difference creates a coarse GrabCut mask, and edges refine it afterwards. These colour-change pixels are definite foreground. A dilation supplies a narrow probable-background margin; everything outside it, including the forced crop border, is definite background.

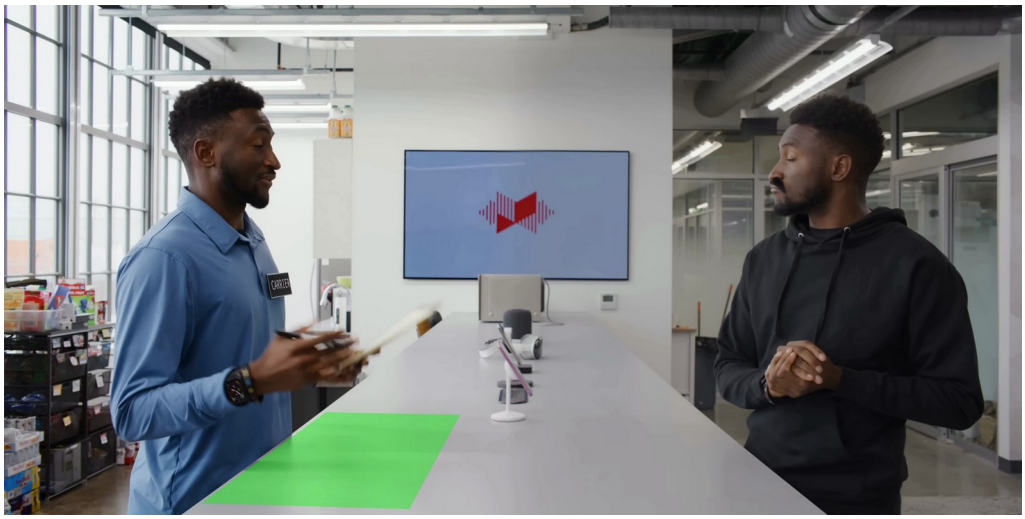
The eroded interior becomes sure foreground, and the band between the eroded and dilated masks is the boundary search area. Canny at 50 and 150, applied to 3 by 3 Gaussian-blurred grayscale, is retained only in this band. Foreground and background markers guide the division of the uncertain band.

Watershed runs only when a coarse foreground, an eroded interior, and candidate boundary edges all exist. Otherwise, the coarse GrabCut result is retained, avoiding complete loss of a thin component simply because erosion removes it. The refined result is restricted to the dilated coarse mask; inner watershed ridges belonging to the coarse foreground are restored.

Because edges are used only near an existing foreground boundary, a static outline cannot independently seed a distant foreground object.


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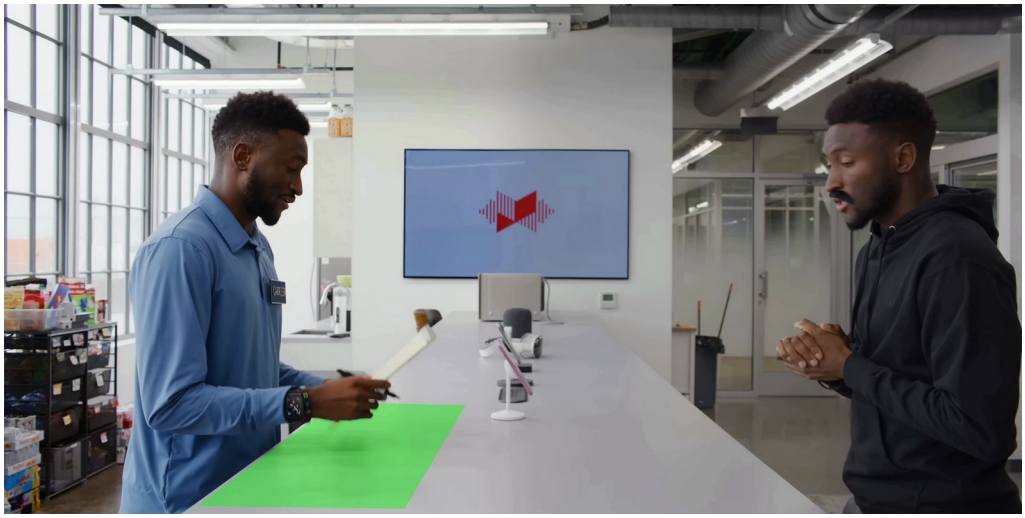

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Version 5

Version 5 expands the context crop according to the selected region. Horizontal padding is the larger of 20 pixels and one quarter of the region width. Vertical padding is the larger of 20 pixels and half the greater region dimension. This gives entering objects more context than the small fixed crops. Unlike Versions 3 and 4, the crop border is not forcibly relabelled as background.

The component filter keeps the largest connected component plus every component intersecting the ad polygon. This is a geometric selection rule, not recognition of the intended object: the largest component can be static background structure, and unrelated detail inside the region is also retained. The broad dilation can bridge narrow neutral gaps and improve continuity, but it expands the silhouette beyond its original boundary.

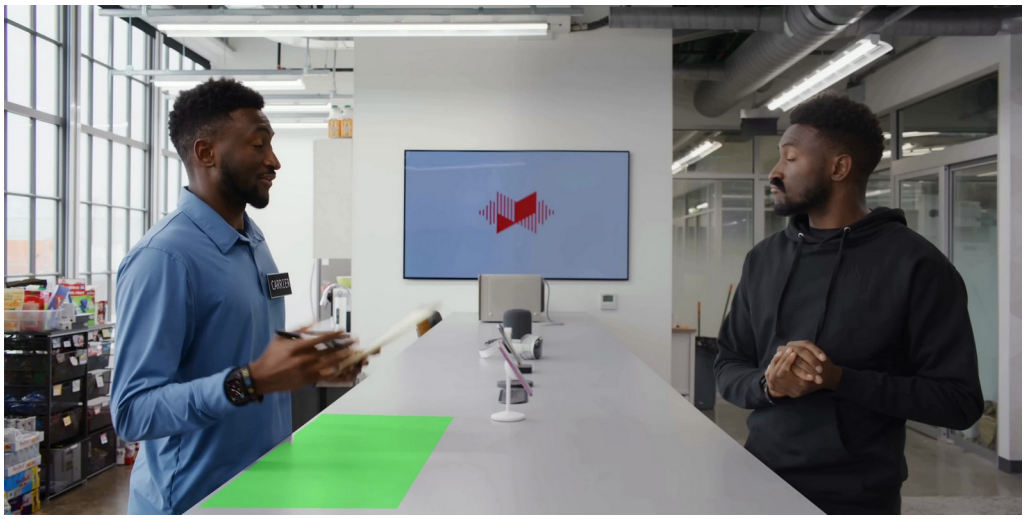
The entire retained silhouette is labelled definite foreground. A 5 by 5 dilated margin is probable background, and the rest is definite background. GrabCut runs three iterations when both classes are available; otherwise the detected foreground is empty. There is no watershed stage. The complete cropped object mask is softened with a 5 by 5 Gaussian blur and inverted, while the caller restricts compositing to the selected polygon.

This seed design prioritises a complete object, but GrabCut cannot shrink away pixels already marked definite foreground. False edges, filled background contours, and excess area introduced by the 13 by 13 dilation can therefore remain locked into the mask. Avoiding a forced background border helps entering objects, yet the larger crop also admits more room detail. The final blur softens these errors rather than correcting their classification.

Those structures are on the surface receiving the overlay, so removing green there is not desired foreground separation. This is consistent with static contours being filled, expanded, and protected as definite foreground. Version 5 therefore trades stronger silhouette continuity for a greater risk of masking background structure; these frames do not support declaring it the most accurate version.


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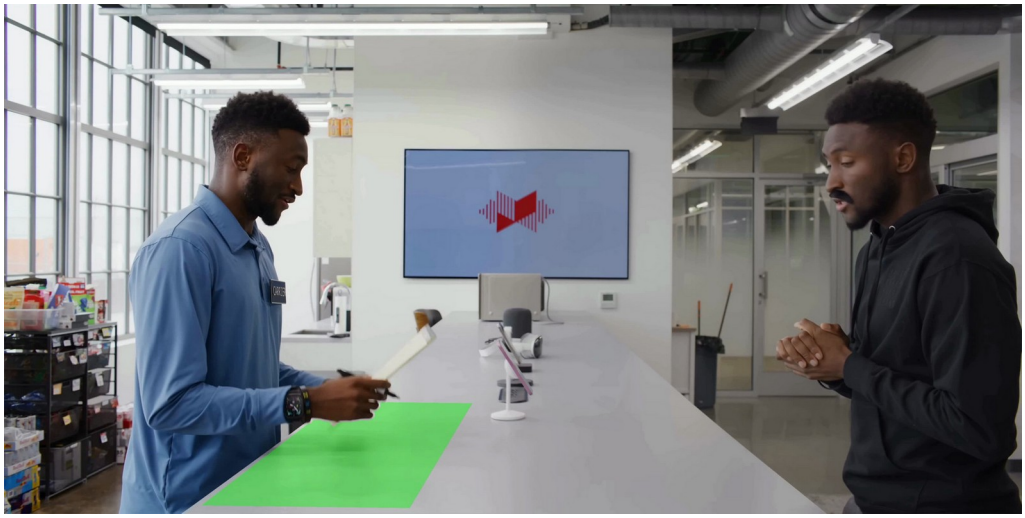

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Current Status

Version 5 is the latest edge-based implementation and combines chroma change, edge evidence, connected components, GrabCut, and a soft boundary. It is the most complete spatial pipeline in the folder, but it has not been shown to be the most accurate. The project does not contain ground-truth masks or a common quantitative evaluation. The per-frame debug displays in the current Version 5 script are commented out.

All methods still depend on a fixed reference and fixed region. Camera movement, a moving ad surface, low-contrast objects, and multiple simultaneous occluders remain open cases. Frame-by-frame methods can also produce visible mask flicker.

Conclusions and Next Steps

The experiments show that preserving the ad on an unchanged surface and keeping a complete object in front of it are separate requirements. A continuous green patch is only a good result when the surface is unobstructed. Green left on a face, hand, or other foreground object is a missed occlusion, while holes over an exposed wall or tabletop are false occlusions. No tested method resolves both errors consistently across the illustrated cases.

The lighting methods reduce the large background removals seen with direct BGR subtraction. Comparing LAB chroma or normalized BGR proportions makes the detector less dependent on brightness, which helps preserve the imran wall and iphone tabletop. The trade-off is weaker separation of objects whose colour resembles the background. The waiter examples make this clear: the first LAB version leaves green on the face, while the more sensitive LAB v2 removes more of that green but also creates scattered holes on the wall. Normalization and small morphological operations improve particular cases, not lighting invariance or complete object recovery.

How edges are admitted matters more than simply adding edge detection. Version 1 combines unrestricted current-frame edges with a zero colour-difference threshold and removes large areas of the tabletop and cabinet. Version 2 preserves these surfaces better by accepting edges only near chroma-change seeds. Version 3 uses filled intensity edges and GrabCut to recover more of the waiter’s face, but also treats the static cabinet decoration as foreground. Edges describe visible structure; they do not establish that the structure is a new object in front of the ad.

Version 4 gives the more conservative result on the illustrated cabinet by restricting edge refinement to an existing colour-seeded boundary, but it cannot recover large regions with no useful colour seeds. Version 5 provides more context and stronger silhouette filling, yet its dilated definite-foreground seeds can lock background detail into the mask. The exposed wall lines and cabinet features demonstrate this cost. A larger foreground mask or a more complex pipeline is therefore not necessarily a more accurate result.

The main design lesson is to improve the reliability of foreground evidence before expanding or smoothing the mask. Versions 2 and 4 are useful baselines for preserving the background; Versions 3 and 5 show the benefits and risks of stronger region completion. This is a qualitative interpretation, not an overall ranking: polygon placement varies, the screenshots are isolated frames, and there are no ground-truth masks or common accuracy and runtime measurements. The next comparison should measure missed foreground, false background removal, boundary quality, flicker, and processing time using identical inputs and saved regions. Fixed-reference and fixed-region assumptions must also be addressed before claiming reliable performance on moving cameras or surfaces.